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🌸 ReLaunchHer: Comprehensive Project Description

🚀 Overview

ReLaunchHer is a high-impact, role-based career re-entry platform specifically designed to empower women returning to the workforce after a career break. Built with a focus on **confidence rebuilding, skill alignment, and opportunity discovery**, the platform provides a guided, personalized journey for every "returnee."

Whether the break was for motherhood, caregiving, health, or relocation, ReLaunchHer transforms that time away into a narrative of **transferable strengths** and provides a data-driven roadmap back to professional success.

[!NOTE] This project follows strict **Software Engineering (SE)** and **Project Management (PM)** standards, making it an ideal case study for academic and professional demonstrations.

Key Objectives

1. **Eliminate the "Gap Penalty":** Reframe career breaks as periods of growth and soft-skill development.
2. **Provide Clear Direction:** Use a multi-factor scoring model to generate personalized 30/60/90-day roadmaps.
3. **Bridge the Skill Gap:** Identify exact missing skills for target roles and provide curated learning paths.
4. **Foster Community:** Connect returnees with mentors who understand their journey and recruiters seeking diverse talent.
5. **Inclusive Access:** Multilingual support to reach women across different linguistic backgrounds.

Core Functionalities

1. Multilingual UI (i18n)

The platform is fully localized in **7 languages**:

- English, Hindi (हिन्दी), Telugu (తెలుగు), Malayalam (മലയാളം), Tamil (தமிழ்), Odia (ଓଡ଼ିଆ), and Bangla (বাংলা).
- **Logic:** Uses a custom `i18n.js` engine that dynamically swaps UI strings based on `data-i18n` attributes, ensuring zero performance lag while providing an inclusive experience.

2. AI-Driven Comeback Planner

The heart of the project is the **Inference Engine** (**logic.js**).

- Users input their background (previous role, break years, reason, target role).
- The engine performs a **Weighted Multi-Factor Analysis** to produce a **Readiness Score**.
- **Outputs:**
 - **Personalized Roadmap:** A phase-by-phase (30/60/90 days) action plan.
 - **Weekly Sprint:** A realistic 5-item checklist tailored to the user's available weekly hours.
 - **Confidence Plan:** Psychological nudges and "STAR" method interview narratives to rebuild self-belief.

3. Role-Based Experiences

The platform adapts its interface for three distinct personas:

- **Returnees:** Focus on planning, upskilling, and tracking progress.
- **Mentors:** Focus on managing mentee requests and providing guidance.
- **Recruiters:** Focus on posting "Returnship" opportunities and identifying high-potential talent.

4. Momentum Dashboard

A central hub where returnees track their progress:

- **Wins Tracker:** Celebrates small victories to maintain momentum.
- **Career Notes:** A space to document learnings and reflections.
- **Job Tracker:** Manage saved and applied "Returnship" programs.

5. Opportunities Hub

A filtered marketplace for:

- **Jobs:** Specifically labeled as "Returnships" or "Hybrid/Remote-friendly."
 - **Mentors:** Industry experts volunteering to guide women back to work.
 - **Learning:** Curated courses mapped directly to the user's skill gaps.
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Behind the Scenes: The Logic Engine

The "magic" of ReLaunchHer lies in its **Readiness Scoring Model**. Unlike simple forms, it uses a complex algorithm to calculate a user's market readiness:

The Scoring Algorithm (**logic.js**)

The total score (0-100) is a weighted sum of:

1. **Experience Depth (20%)**: Years of previous professional work.
2. **Break Penalty Mitigation (-20% to 0%)**: Instead of a flat penalty, the engine *reduces* the penalty if the user was upskilling or freelancing.
3. **Skill Match Ratio (22%)**: How many "Required Skills" the user already possesses for their target role.
4. **Confidence & Support (16%)**: Factors in current confidence levels and the presence of a support system.
5. **Education & Leadership (16%)**: Academic background and previous management experience.
6. **Engagement (10%)**: Weekly hours committed and networking activity.

Transferable Strengths Model

The engine maps "Break Reasons" to professional skills:

- **Motherhood** → "Structured planning under uncertainty" & "Stakeholder management."
- **Caregiving** → "Crisis management" & "Emotional intelligence."
- **Relocation** → "Adaptability" & "Cross-cultural communication."



Tech Stack & Architecture

Frontend

- **Languages:** HTML5, CSS3, Vanilla JavaScript.
- **Styling:** Custom CSS with a focus on "Modern Premium Aesthetics" (Gradients, Glassmorphism, Micro-animations).
- **State Management:** `storage.js` handles client-side caching and API interaction.

Backend

- **Server:** Node.js (HTTP Module) — a lightweight, fast, and secure custom server.
- **Database:** SQLite with the `better-sqlite3` library (high-performance, file-based persistence).
- **Session Management:** Secure cookie-based sessions for role-based authentication.

Database Schema

- `users`: Core profile data (Role, Language, Credentials).
- `plans`: JSON-serialized career plans and AI analysis results.
- `progress`: Tracking checklists, wins, and notes.
- `job_states`: Relational mapping of saved and applied jobs.
- `posted_jobs` & `mentor_profiles`: Data for the opportunities hub.



Software Engineering Artifacts

The project is documented using industry-standard modeling techniques to ensure structural integrity:

- **Data Flow Diagrams (DFD):** Level 0 and Level 1 diagrams mapping the flow of profile data through the inference engine.
- **UML Class Diagrams:** Demonstrating the inheritance and relationships between `User`, `Returnee`, `Mentor`, and `Recruiter`.
- **Sequence Diagrams:** Visualizing the asynchronous interaction between the UI Controller, the Logic Engine, and the SQLite Database.
- **State Chart Diagrams:** Tracking the lifecycle of a job application from "Discovered" to "Accepted for Interview."



Security & Performance

- **Secure Sessions:** Custom session management using server-side encrypted cookies and manual token pruning.
- **Security Headers:** Implementation of `CSP`, `X-Frame-Options`, and `nosniff` to protect against XSS and Clickjacking.
- **Performance:** Optimized static file serving with zero-cache headers for the dev environment and standard MIME type mapping.



Testing & Quality Assurance

The project includes a robust automated test suite (`tests/analysis.test.js`) using the **Node.js Native Test Runner**:

- **Logic Validation:** Ensures that the scoring algorithm reacts correctly to different career break lengths.
- **Persona Verification:** Validates that the "Returning Mother" and "Career Switcher" personas receive distinct, tailored advice.
- **Regression Testing:** Prevents changes in logic from breaking the dynamic roadmap builder.



UI/UX & Design Aesthetics

- **Glossy Design System:** Uses CSS variables for a curated, premium color palette.
 - **Glassmorphism:** Implementing blurred backgrounds and semi-transparent layers for a modern look.
 - **Micro-animations:** Subtle hover transitions and progress bars to enhance the "Confidence" experience.
 - **Responsive Layouts:** Mobile-first approach ensuring accessibility across all devices.
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Presentation Strategy (For 100+ People)



The "Hook"

Start with the "Career Gap" problem. Millions of women leave the workforce for family reasons and struggle to return because of outdated resumes or lost confidence. ReLaunchHer is the bridge.



The Demo Flow

1. **Landing:** Show the stunning, multilingual UI. Switch from English to Hindi to demonstrate inclusivity.
2. **Planner:** Fill out a "Returnee" profile (e.g., Sarah, Software Engineer, 4-year break).
3. **Inference:** Click "Generate Plan." Show the 30/60/90-day roadmap. Explain that this isn't random—it's calculated based on her specific skills and gaps.
4. **Dashboard:** Show how the plan is saved. Add a "Win" (e.g., "Completed React module") to show the momentum.
5. **Recruiter View:** Briefly mention that recruiters see Sarah not as someone with a gap, but as a "High Readiness" candidate.



The "Wow" Factor

Focus on the **Readiness Score**. Explain that we are using logic to **quantify potential**, making it easier for both the woman to feel confident and the recruiter to hire.

ReLaunchHer isn't just a job board; it's a **Confidence Engine** that turns "I've been away" into "I'm ready to lead."